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Joint Target-Speaker ASR and Activity Detection

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Abstract

Target-speaker automatic speech recognition (TS-ASR) has shown promise in transcribing speech in multi-speaker scenarios by focusing on a specific speaker. However, existing approaches employ a cascaded design, where voice activity detection (VAD) and TS-ASR are optimized separately. This separation leads to unstable training and error accumulation, limiting overall performance. We address these issues by proposing TS-ASR-AD, a joint end-to-end model that integrates VAD with TS-ASR, enabling stable training and reducing error accumulation. Moreover, improved training stability leads to better connectionist temporal classification (CTC) alignment in token probabilities, further enhancing transcription accuracy. Our approach outperforms previous studies and achieves a word error rate (WER) of 6.61% and 14.81%, and a diarization error rate (DER) of 1.23% and 2.66% on the Libri2Mix and Libri3Mix datasets, respectively.

Index Terms: multi-talker ASR, target-speaker ASR

1. Introduction

In multi-speaker environments such as group conversations and meetings, humans effortlessly distinguish between speakers, track speech activity, and rapidly comprehend spoken content. This capability remains challenging for automatic systems. Previous studies have primarily concentrated on methods e.g., speaker diarization (SD) [1, 2], speech separation (SS) [3], and automatic speech recognition (ASR) to automate these tasks. SD determines the speech activities of all speakers in an input audio mixture, answering the question of "who spoke when." SS extracts the speech signals of individual speakers from a mixture, while ASR transcribes segmented speech signals. In multi-speaker scenarios, combining SS and ASR enables the extraction of "who spoke what." Numerous studies have proposed cascaded pipelines [4, 5] that integrate these tasks, achieving reasonable performance.

Nevertheless, error propagation through independently trained models limits the performance of these cascaded approaches. Prior works have explored joint modeling to address this limitation. Specifically, they model SD and SS [6–8], SD and ASR [9, 10], SS and ASR [11, 12], or even all three tasks jointly [13, 14]. These joint modeling strategies mitigate error propagation and enhance performance by leveraging shared knowledge between tasks. However, these approaches often employ a fixed number of speakers or show suboptimal performance to reliably associate speakers with their speech.

Unlike SD-based approaches, which can process multiple speakers simultaneously, target speaker (TS)-based approaches identify and process the speech of a specific target speaker, making them inherently adaptable to an unknown number of speakers. Prior research has explored target speaker-based voice activity detection (TS-VAD) [15, 16] for determining target speaker activity, TS-SS [17, 18] for extracting target speaker speech, and TS-ASR [19, 20] for transcribing target speech. These methods generalize better to varying speaker counts than SD-based models by focusing on individual speakers. However, they often suffer from unstable training dynamics due to the dependence on target speaker embeddings, which can vary significantly across datasets and conditions. Additionally, applying conventional TS-ASR models with recurrent neural network transducer (RNN-T)-based decoder [20] or a connectionist temporal classification (CTC)-based decoder can lead to misalignment issues due to the high number of blank tokens in speech mixtures. These factors collectively contribute to training instability, affecting both convergence and generalization.

Existing TS-based methods typically rely on external speaker cues for network conditioning. Most prior works have used speaker embeddings [19, 21], while others have explored embedding-free [22] or multimodal approaches [23, 24]. Additionally, many methods depend on computationally intensive training strategies such as permutation invariant training (PIT) [25] or attention-based decoders [26], leading to increased computational complexity. We overcome these challenges by introducing a joint target-speaker ASR and activity detection (TS-ASR-AD) model. This model integrates ASR and VAD in an end-to-end manner and relies on a computationally inexpensive lightweight CTC decoder. A key challenge of applying CTC to speech mixtures is the presence of numerous blank tokens due to non-segmented speech or interference from other speakers. We hypothesize that our framework stabilizes training by providing additional speech activity supervision, thereby enhancing the regularization of CTC token probabilities. Unlike [19], which applies joint speech modeling for fusion before a single CTC decoding step, our method explicitly leverages VAD during training, bringing increased stability and better alignment in CTC token probabilities. Our key contributions are:

- We propose TS-ASR-AD, a novel joint model that integrates VAD and TS-ASR, addressing the instability and error accumulation issues caused by separately optimized systems.
- We highlight that integrating VAD with ASR stabilizes training and regularizes CTC token probabilities. Thereby mitigating the blank token accumulation observed in CTC alignment for speech mixtures.
- We demonstrate through extensive experiments on the LibriMix [27] and LibriSpeechMix [28] datasets that our method outperforms previous studies in detecting target speech activity and transcribing target speaker speech.

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2. Related work

Previous cascaded approaches often relied on SD or TS-VAD as the initial step for group conversation or meeting-like multi-speaker transcription [4,5]. The introduction of end-to-end systems for SD improved performance, particularly in detecting speaker changes in highly overlapped scenarios [29,30]. In contrast, TS-VAD [15] used speaker embeddings to predict each speaker's activity, achieving solid results in the CHiME-6 challenge [31], but it was not integrated with subsequent ASR systems, limiting performance in downstream tasks.

Multi-speaker ASR typically serves as the second and final step in such cascaded systems. Early multi-speaker ASR work involved modular setups using speech separation followed by traditional single-speaker ASR systems, which benefited from powerful single-speaker ASR models [32]. However, these cascading systems had limited performance due to the distortions caused by the separation models [33]. To address this, methods emerged that transcribe the speech of all speakers without relying on speech separation, using techniques like permutation invariant training (PIT) [25] and serialized output training (SOT) [28], although these approaches remained limited in terms of speaker identification. Meanwhile, Speaker-Attributed ASR (SA-ASR) [34] utilized speaker embeddings to enhance transcriptions, especially in highly overlapped scenarios. TS-ASR [19], a variant of SA-ASR, focused specifically on transcribing the target speaker's speech. However, despite these advances, such models are often computationally expensive and lack awareness of speaker activities within the speech mixture.

3. Proposed method

We introduce TS-ASR-AD, a novel joint end-to-end model that directly integrates VAD with TS-ASR, tackling the instability and error accumulation issues caused by separately optimized systems. An overview of the proposed framework is shown in Figure 1, which consists of four key components: 1) an upstream embedder module that generates the target speaker embedding from its profile, 2) an adaptation module that utilizes the target speaker embedding to adjust the input mixture features, 3) an upstream encoder module that enhances the adapted features for all tasks, and 4) downstream decoder modules that produce the task-specific predictions.

3.1. Embedder

The embedder's role is to encode the input target speaker cue into a representative vector, known as an embedding. Following prior work [19], we use speaker profiles (i.e., a few seconds of clean, segmented speech from the speaker) as the input cues. Given a target speaker cue C_{s} the embedder generates an embedding E_{s} as follows:

$$E_{\text{s}} = Embedder(C_{\text{s}}) \tag{1}$$

where $C_{\text{s}} \in \mathbb{R}^{1\times T}$ and $E_{\text{s}} \in \mathbb{R}^{1\times D_{\text{emb}}}$ with T the audio time sample size and D_{emb} the embedder's hidden size.

3.2. Adaptation Module and Encoder

The adaptation module's role is to align the audio mixture with the embedding. Inspired by previous work [19, 21], we use Feature-wise Linear Modulation (FiLM) [35] for the adaptation process. Additionally, we leverage Self-Supervised Learning (SSL) techniques for the encoder, as they have shown strong performance in downstream speech processing tasks [36]. Given the audio mixture X and the embedding E_{s}, features F are first extracted from X using the encoder's feature extraction method (e.g., log-Mel spectrograms, CNN representations, etc.). These extracted features H are then adapted to the embedding E_{s} through the FiLM module. The adapted speaker features H_{s} are defined as:

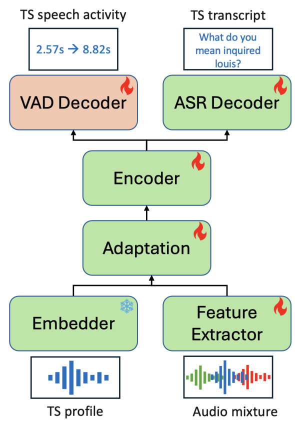

Figure 1: *Illustration of our joint target speaker activity detection model (TS-ASR-AD).*

$$\begin{aligned} H_s &= Encoder(X, E_{\text{s}}) \\ &= Encoder(FiLM(H, E_{\text{s}})) \\ &= Encoder(FiLM(Extractor(X), E_{\text{s}})) \end{aligned} \tag{2}$$

$$FiLM(X, E_{\text{s}}) = \gamma(E_{\text{s}}) \cdot X + \beta(E_{\text{s}}) \tag{3}$$

where $X \in \mathbb{R}^{1\times T}$, $H \in \mathbb{R}^{1\times F\times D_{\text{enc}}}$, and $H_{\text{s}} \in \mathbb{R}^{1\times F\times D_{\text{enc}}}$ with F the feature size and D_{enc} the encoder hidden size.

3.3. Decoders

The decoders are responsible for generating predictions. Since the proposed method outputs results for two tasks, we use two decoders: the ASR decoder and the VAD decoder. Given the adapted speaker features H_{s}, the ASR predictions $\hat{Y}_{\text{asr}}$ and VAD predictions $\hat{Y}_{\text{vad}}$ are defined as:

$$\hat{Y}_{\text{asr}} = ASR(H_{\text{s}}) \tag{4}$$

$$\hat{Y}_{\text{vad}} = VAD(H_{\text{s}}) \tag{5}$$

where $\hat{Y}_{\text{asr}} \in \mathbb{R}^{1\times F\times V}$ and $\hat{Y}_{\text{vad}} \in \mathbb{R}^{1\times F\times C}$ with V the vocabulary size for ASR and C the class size for VAD.

3.4. Training objective

The proposed model is jointly optimized for two tasks. The system loss L is a combination of the ASR loss L_{asr} and the VAD loss L_{vad}. Following previous work [19], we use the CTC loss [37] for the ASR loss. For the VAD loss, we apply the multiclass classification Cross Entropy loss. Given the reference speaker tokens Y_{asr} and reference speaker activity Y_{vad}, the system loss L is defined as:

$$\begin{aligned} L &= \lambda_{\text{asr}} \cdot L_{\text{asr}} + \lambda_{\text{vad}} \cdot L_{\text{vad}} \\ &= \lambda_{\text{asr}} \cdot CTC(Y_{\text{asr}}, \hat{Y}_{\text{asr}}) + \lambda_{\text{vad}} \cdot CE(Y_{\text{vad}}, \hat{Y}_{\text{vad}}) \end{aligned} \tag{6}$$

where $Y_{\text{asr}} \in \mathbb{N}^{1\times L}$ and $Y_{\text{vad}} \in \mathbb{N}^{1\times F}$ with L the tokens length size.

Table 1: *ASR results presented in word error rate (WER). The proposed method uses WavLM Base+. "-" indicates that the result is not available. Best results are shown in* ***bold***.

Method	Params	Hours	LibriMix		LibriSpeechMix		
			2-Mix	3-Mix	1-Mix	2-Mix	3-Mix
Baselines							
Whisper-small-SS-TTI [26]	242M	100h	15.75	-	-	-	-
TS-ASR (reproduced)	96M	100h	15.59	30.10	4.54	18.70	29.85
WavLM + TSE [19]	96M	100h	13.84	-	-	-	-
Whisper-small-SS-TTI [26]	242M	460h	11.81	30.52	-	-	-
Whisper-small-SS-TTI [26]	242M	960h	-	-	-	8.89	15.85
Transformer SA-ASR [34]	unk.	960h	-	-	3.90	**6.40**	**8.50**
Proposed							
TS-ASR-AD	97M	100h	**12.36**	**29.85**	**3.35**	**10.64**	**20.58**
TS-ASR-AD	97M	460h	**6.61**	**14.81**	**3.01**	7.92	15.97
TS-ASR-AD	319M	960h	-	-	**2.75**	6.61	13.17

4. Experiments

4.1. Datasets and Metrics

We evaluated the proposed method on the LibriMix [27] and LibriSpeechMix [28] datasets. LibriMix consists of speech mixtures with complete overlap, derived from LibriSpeech [38] and WHAM!'s noises [39]. It is commonly used for speaker diarization, speech separation, and speech recognition tasks. The dataset includes subsets with 2 or 3 speakers. Since the original dataset does not provide speaker profile information, we followed [19] to collect speaker enrollments, using only 5 seconds of audio instead of the original 15 seconds. We evaluated our method on the 2-Mix and 3-Mix using clean mixtures in max mode.

LibriSpeechMix [28], on the other hand, consists of speech mixtures with partial overlap, also derived from LibriSpeech [38]. While primarily intended for speech recognition, LibriSpeechMix can be extended to other tasks. Since the original dataset does not provide a training set, we followed [28] to generate the training set dynamically. For evaluation, we only used the first speaker profile segment from the available segments. We tested our method on the 1-Mix, 2-Mix, and 3-Mix subsets.

ASR results are assessed using the Word Error Rate (WER), a standard metric for evaluating ASR systems. For methods where speaker estimation is involved, we may use concatenated minimum-permutation WER (cp-WER) [31] to account for potential speaker mismatches. On the other hand, VAD results are evaluated using the diarization error rate (DER), a widely used metric for multi-speaker activity detection. DER is computed using the meeteval[1] toolkit. Following prior work [40], DER is calculated with a median filter of 11 and 0s of collar.

4.2. Model Settings

Inspired by previous work [19, 21], the upstream embedder follows the ECAPA-TDDN [41] deep network architecture and is instantiated with a checkpoint available here[2] trained on VoxCeleb [42]. The upstream encoder follows the WavLM [36] architecture and is instantiated with checkpoints available here[3]. The adaptation module follows the FiLM [35] architecture and is trained from scratch. Finally, the ASR decoder consists a dense followed by a projection linear layer with sizes $D_{\text{enc}} \times D_{\text{enc}}$ and $D_{\text{enc}} \times V$ respectively where $V = 31$. Similarly, the VAD decoder a dense followed by a projection linear layer with sizes $D_{\text{enc}} \times D_{\text{enc}}$ and $D_{\text{enc}} \times C$ where $C = 2$.

Models are trained for a total of 50k, 100k, or 500k steps-depending on the training set size and with a batch size of 32. The embedder is kept frozen for the entire duration of the training while the encoder is only kept frozen for the first 10k steps. The model is optimized with Adam [43] with a learning rate of $1 \cdot 10^{-4}$. Warmup occurs for the first 5k steps and linear decay is introduced after reaching half of the total steps.

[1]https://github.com/fgnt/meeteval

[2]https://huggingface.co/speechbrain/spkrec-ecapa-voxceleb

[3]https://huggingface.co/microsoft/wavlm-base-plus

5. Results and Discussions

5.1. ASR and VAD results

We evaluated the proposed TS-ASR-AD model on two datasets, LibriMix and LibriSpeechMix, which feature different overlap settings. The results for ASR are presented in Table 1, while the VAD results are shown in Table 2.

For ASR, TS-ASR-AD consistently outperforms TS-ASR under identical training conditions. Specifically, it improves from 15.59% to 12.36% on the Libri2Mix evaluation set and from 30.10% to 29.85% on the Libri3Mix set. These improvements highlight the benefits of joint ASR and VAD training in multi-speaker scenarios. With additional data, TS-ASR-AD achieves a WER of 6.61% and 14.81% on the Libri2Mix and Libri3Mix sets, outperforming previous studies. On the LibriSpeechMix dataset, TS-ASR-AD outperforms Whisper-small-SS-TTI [26] and surpasses Transformer SA-ASR [34] on the 1-Mix set, while delivering comparable performance on the 2-Mix set, despite Transformer SA-ASR being trained with significantly more resources.

For VAD, TS-ASR-AD outperforms TS-VAD on Libri2Mix, improving from 3.33% to 2.87%. On Libri3Mix, while TS-ASR-AD's performance is still below that of TS-VAD, it outperforms previous work [40]. With additional data, TS-ASR-AD achieves a DER of 1.23% and 2.66% on the Libri2Mix and Libri3Mix sets, outperforming previous studies. Moreover, as the first to report VAD results on the LibriSpeechMix dataset, we establish new baseline DER values of 0.14%, 2.79%, and 5.38% for the 1-Mix, 2-Mix, and 3-Mix sets, respectively.

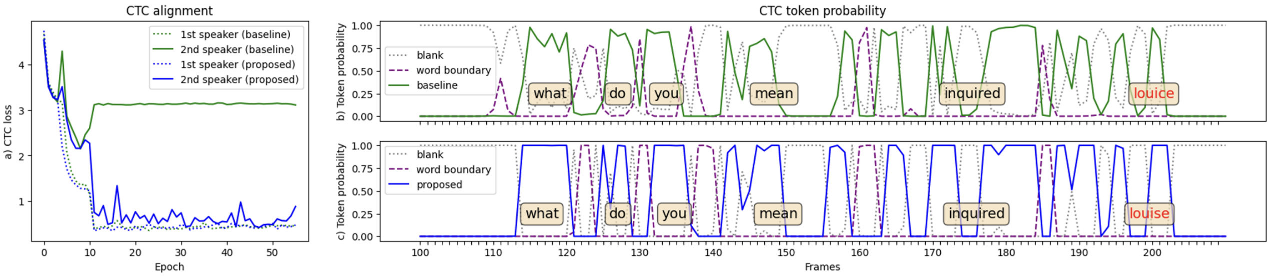

Figure 2: *Left (a): CTC loss for the 1st and 2nd speaker. The baseline is only trained on ASR while the proposed method is jointly trained on ASR and VAD. Right (b+c): CTC token probability when transcribing the 2nd speaker. Token probabilities are smoother for the proposed method. The reference transcript was: "what do you mean inquired louis".*

Table 2: *VAD results presented in diarization error rate (DER). If not specified, the training dataset is LibriMix-100h or LibriSpeechMix-100h and the proposed method uses WavLM Base+. "-" indicates that the result is not available.*

	LibriMix			LibriSpeechMix		
Method	**2-Mix**	**3-Mix**	**2&3-Mix**	**1-Mix**	**2-Mix**	**3-Mix**
Baselines						
WavLM + SD [36]	3.50	-	-	-	-	-
TS-VAD [40]	-	-	7.28	-	-	-
USED [40]	-	-	5.20	-	-	-
TS-VAD (reproduced)	3.33	6.58	4.96	-	-	-
Proposed						
TS-ASR-AD	2.87	7.14	5.01	0.22	3.80	8.30
TS-ASR-AD ($460h$)	**1.23**	**2.66**	**1.95**	0.19	3.21	6.49
TS-ASR-AD ($960h, large$)	-	-	-	**0.14**	**2.79**	**5.38**

5.2. CTC alignment for speech mixtures

As shown in Section 5.1, the proposed method consistently improves performance in multi-speaker scenarios. We further hypothesize that CTC alignment is degraded in speech mixtures, particularly due to 'delayed audio', which results in an increase of blank tokens in the CTC hypothesis. To demonstrate this, we conducted a simple experiment where we transcribe the speech of the first or last speaker in a two-speaker mixture. The results are shown in Figure 2.

When the model is optimized solely for ASR (baseline), we observe that CTC struggles to converge when transcribing the last speaker, but not the first. This confirms that CTC alignment becomes more challenging for delayed or unsegmented audio. However, when the model is jointly optimized for ASR and VAD (proposed), this issue is alleviated, and CTC converges in both cases. Additionally, we note that the token probabilities for CTC are less stable when VAD is not optimized alongside ASR. In conclusion, jointly optimizing ASR and VAD significantly improves CTC alignment, which likely contributes to the superior performance of the proposed method.

5.3. Effect of multi-task weight

Table 3 presents the ASR and VAD results for various multi-task weight values, λ_{asr}. We find that setting λ_{asr} too high (≥ 0.9) does not yield the best performance in multi-task learning. We hypothesize that this is related to the CTC alignment issues discussed in Section 5.2. From the empirical results, we observe that the optimal value for λ_{asr} is 0.7, which results in a WER of 12.36% and a DER of 2.87%. This value was used in the proposed method.

Table 3: *ASR and VAD results for different multi-task weight. The dataset used is Libri2Mix-100h and the proposed method uses WavLM Base+. Metric is WER for ASR and DER for VAD.*

Method	**Weight**	**ASR**	**VAD**
TS-ASR-AD	$\lambda_{\text{asr}} = 0.3$	13.99	3.41
	$\lambda_{\text{asr}} = 0.5$	12.89	3.02
	$\lambda_{\text{asr}} = 0.7$	**12.36**	**2.87**
	$\lambda_{\text{asr}} = 0.9$	15.52	4.26

5.4. Effect of enrollment duration

Table 4: *ASR and VAD results with different enrollment duration. The dataset is Libri2Mix-100h and the proposed method uses WavLM Base+. Metric is WER for ASR and DER for VAD.*

Method	**Duration**	**ASR**	**VAD**
TS-ASR-AD	3s	15.57	3.79
	5s	12.36	2.87
	10s	11.11	2.43
	15s	**10.36**	**2.17**

Table 4 presents the ASR and VAD results for different enrollment durations. We observe that performance improves with longer enrollments, which is expected since the embedder was kept frozen. When the enrollment duration is tripled from 5 to 15 seconds, ASR improves by 2% and DER by 0.6%. The performance gap is particularly noticeable with enrollments shorter than 5 seconds. We hypothesize that 5 seconds is the minimum duration at which the proposed method delivers strong performance, which is why we use this value for the proposed method.

6. Conclusion

In this paper, we introduced a novel methodology that seamlessly integrates ASR and VAD for TS scenarios. By adding a VAD decoder to the TS-ASR architecture, we proposed a joint model that addresses the instability and error accumulation issues caused by separately optimized systems. Extensive experiments demonstrate that our approach improves CTC alignment in token probabilities for speech mixtures and outperforms previous studies on the LibriMix and LibriSpeechMix datasets, for both ASR and VAD tasks. Furthermore, we established a new baseline for the VAD task on the LibriSpeechMix dataset, being the first to report these results.

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